

Engine Does Not Shut Down When Vessel Shutdown 1 Signal Active on Shutdown Unit

Symptoms

- The SDU410 does **not** shut down the engine with signal 1 active at OEM X6 connection.

How To Use This Tree

This symptom tree can be used to troubleshoot an OEM circuit malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The SDU410 input signals are switches. These switches are normally open and closed to activate a shutdown.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check customer interface box	
	STEP 1A. Check the customer interface box logic unit LED illumination.	Are any alarms active or LEDs illuminated?
	STEP 1B. Check the SDU410 power supply wire for +24-VDC.	Less than +24-VDC?
STEP 2.	Check customer interface box wiring	
	STEP 2A. Check the signal 1 signal and return wires for an open at the X6 connection.	Less than 10 ohms?

STEP 1. Check customer interface box

STEP 1A. Check the customer interface box logic unit LED illumination.

Conditions :

- Check the DCU410 unit for alarms and LED illumination.

Action	Specification/Repair	Next Step
Check for alarms and LED illumination on the DCU410 unit.	Are any alarms active or LEDs illuminated? YES	Contact a Cummins® Authorized Repair Location
	Are any alarms active or LEDs illuminated? NO	1B

STEP 1B. Check the DCU410 power supply wire for +24-VDC.**Conditions :**

- Open the customer interface box

Action	Specification/Repair	Next Step
Check the voltage at the shutdown unit supply 24-VDC at the SDU410 unit. • Place one test lead on the shutdown unit supply 24-VDC supply wire at the SDU410 unit. • Place the other test lead on the shutdown unit return wire at the SDU410 unit. Refer to the appropriate circuit diagram or wiring diagram for connection pin identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	Repair complete
	Less than +24-VDC? NO	2A

STEP 2. Check customer interface box wiring**STEP 2A. Check the signal 1 signal and return wires for an open at the X6 connection.**

Conditions :

- Open the customer interface box
- Disconnect the signal 1 signal and return wires on the OEM X6 connection.

Action	Specification/Repair	Next Step
Check the signal 1 signal and return wires for an open at the X6 connection. • Place one test lead on the signal 1 signal wire at the X6 connector. • Place the other test lead on the signal 1 signal wire at the OEM connection. • Place one test lead on the signal 1 return wire at the X6 connector. • Place the other test lead on the signal 1 return wire at the OEM connection. Refer to the appropriate circuit diagram or wiring diagram for connection pin identification.	Less than 10 ohms? YES	Repair complete
	Less than 10 ohms? NO Repair : Replace the wire. Refer to the OEM installation instructions.	Contact a Cummins® Authorized Repair Location.